

Chapter 5: Enumeration (Week 8 & 9)

Enumeration Technical Report

Target: Metasploitable 2 | Attacker: Kali Linux

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1.0 Introduction

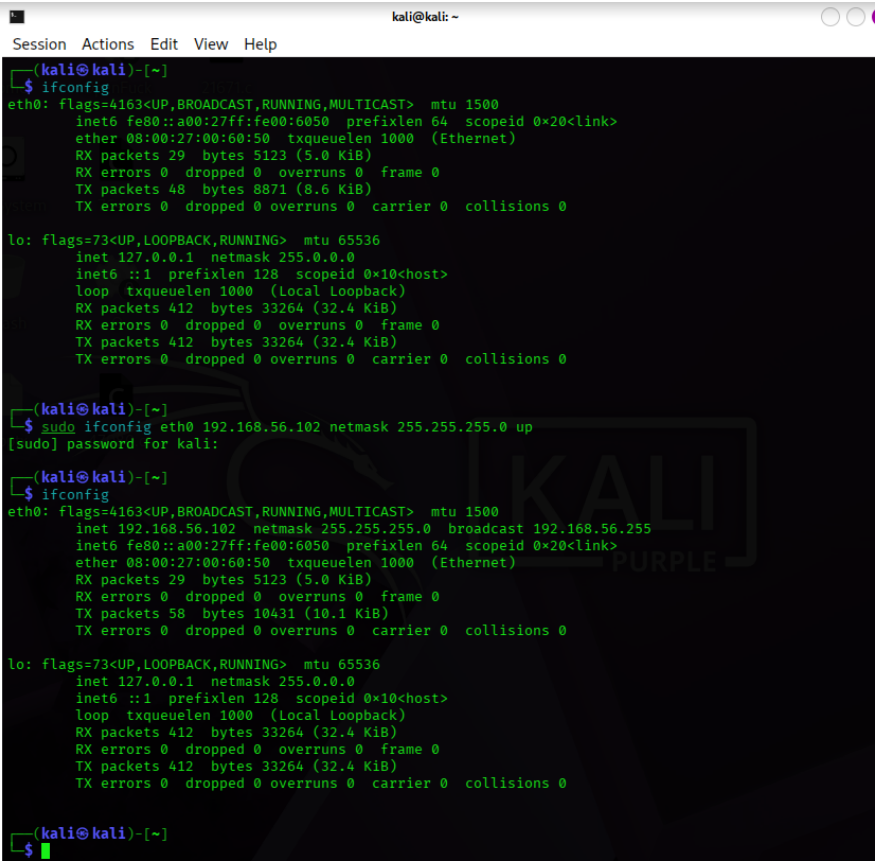
Enumeration is the active phase of a penetration test in which the tester connects directly to a target to extract detailed information such as open ports, running services and their versions, user accounts, shared resources and the operating system. This information forms the foundation for the exploitation phase, because it turns general reconnaissance into concrete, actionable attack paths.

The objective of this lab is to perform a range of enumeration techniques against a single vulnerable target and to interpret the results to identify potential weaknesses. As required by the lab brief, at least ten enumeration challenges were completed, covering port discovery, service and OS fingerprinting, FTP, SMB, NFS/RPC and SMTP enumeration.

2.0 Lab Environment

The lab was conducted in an isolated VirtualBox Host-Only network. Kali Linux was used as the attacker machine and Metasploitable 2 as the deliberately vulnerable victim machine. The IP addresses assigned to each machine are shown below.

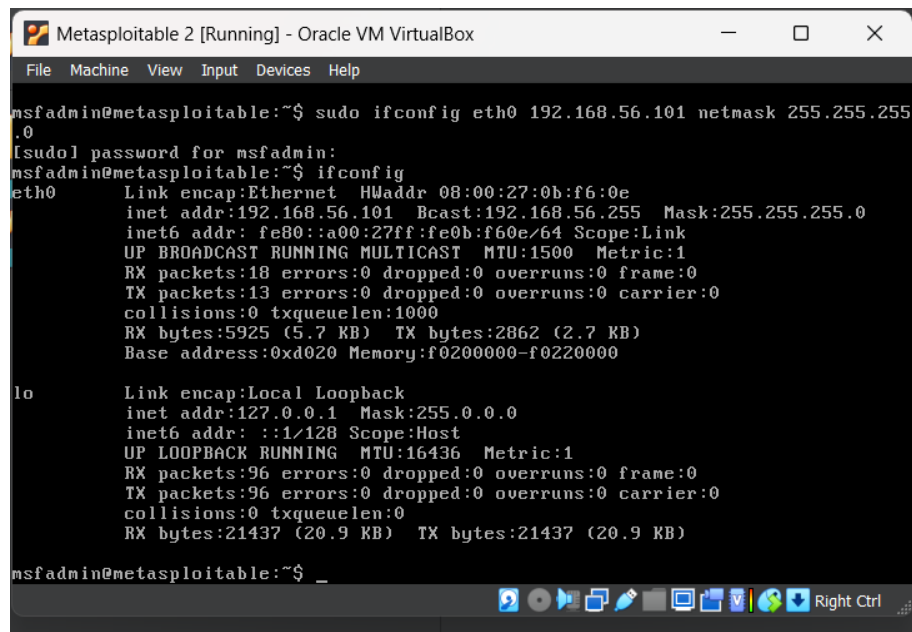
Attacker (Kali Linux): 192.168.56.102



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet6 fe80::a00:27ff:fe00:6050 prefixlen 64 scopeid 0x20<link>  
    ether 08:00:27:00:60:50 txqueuelen 1000 (Ethernet)  
    RX packets 29 bytes 5123 (5.0 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 48 bytes 8871 (8.6 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 412 bytes 33264 (32.4 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 412 bytes 33264 (32.4 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
(kali@kali)-[~]  
$ sudo ifconfig eth0 192.168.56.102 netmask 255.255.255.0 up  
[sudo] password for kali:  
(kali@kali)-[~]  
$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.56.102 netmask 255.255.255.0 broadcast 192.168.56.255  
    inet6 fe80::a00:27ff:fe00:6050 prefixlen 64 scopeid 0x20<link>  
    ether 08:00:27:00:60:50 txqueuelen 1000 (Ethernet)  
    RX packets 29 bytes 5123 (5.0 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 58 bytes 10431 (10.1 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 412 bytes 33264 (32.4 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 412 bytes 33264 (32.4 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
(kali@kali)-[~]  
$
```

Figure 1: Attacker machine (Kali Linux) network configuration

Victim (Metasploitable 2): 192.168.56.101

A screenshot of a terminal window titled "Metasploitable 2 [Running] - Oracle VM VirtualBox". The terminal shows a user prompt "msfadmin@metasploitable:~\$" followed by the command "sudo ifconfig eth0 192.168.56.101 netmask 255.255.255.0". The output shows the configuration for the "eth0" interface, including its IP address, broadcast address, netmask, and various statistics. The prompt then changes to "msfadmin@metasploitable:~\$" after the command is executed. The terminal window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". The bottom of the window shows a taskbar with various icons and a "Right Ctrl" button.

```
msfadmin@metasploitable:~$ sudo ifconfig eth0 192.168.56.101 netmask 255.255.255.0
[sudo] password for msfadmin:
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:0b:f6:0e
          inet addr:192.168.56.101  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe0b:f60e/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:18 errors:0 dropped:0 overruns:0 frame:0
          TX packets:13 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:5925 (5.7 KB)  TX bytes:2862 (2.7 KB)
          Base address:0xd020  Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:96 errors:0 dropped:0 overruns:0 frame:0
          TX packets:96 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:21437 (20.9 KB)  TX bytes:21437 (20.9 KB)

msfadmin@metasploitable:~$ _
```

Figure 2: Victim machine (Metasploitable 2) network configuration

3.0 Enumeration Challenges

Challenge 2 — Fast Nmap Scan

Command

```
nmap -F 192.168.56.101
```

Explanation

The -F (fast) option scans the 100 most common TCP ports instead of all 65,535, giving a quick overview of which services are exposed. This is usually the first step of active enumeration to map the target attack surface.

```
(kali@kali)-[~]
$ nmap -F 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-07 00:56 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns
or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.101
Host is up (0.00016s latency).
Not shown: 82 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
513/tcp   open  login
514/tcp   open  shell
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
8009/tcp  open  ajp13
MAC Address: 08:00:27:0B:F6:0E (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.28 seconds

(kali@kali)-[~]
$
```

Figure 3: Open ports discovered on the target via a fast Nmap scan

Findings

The target exposes a large number of services, including FTP (21), SSH (22), Telnet (23), SMTP (25), DNS (53), HTTP (80), RPCbind (111), NetBIOS/SMB (139/445), NFS (2049), MySQL (3306), PostgreSQL (5432) and VNC (5900). This indicates a heavily exposed Linux host with many potential entry points.

Challenge 16 — Service Version Detection

Command

```
nmap -sV 192.168.56.101
```

Explanation

The -sV flag performs service and version detection by probing each open port and matching the responses against Nmap's signature database. Knowing the exact software versions lets an attacker look up known vulnerabilities (CVEs) tied to those versions.

```
kali@kali: ~
Session Actions Edit View Help
6000/tcp open  X11
8009/tcp open  ajp13
MAC Address: 08:00:27:0B:F6:0E (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.28 seconds

(kali@kali)-[~]
$ nmap -sV 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-07 00:58 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns
s or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.101
Host is up (0.00011s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi      GNU Classpath grmiregistry
1524/tcp  open  bindshell     Metasploitable root shell
2049/tcp  open  nfs           2-4 (RPC #100003)
2121/tcp  open  ftp           ProFTPD 1.3.1
3306/tcp  open  mysql         MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql    PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc           VNC (protocol 3.3)
6000/tcp  open  X11           (access denied)
6667/tcp  open  irc           UnrealIRCd
8009/tcp  open  ajp13         Apache Jserv (Protocol v1.3)
8180/tcp  open  http          Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:0B:F6:0E (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.78 seconds

(kali@kali)-[~]
$
```

Figure 4: Service and version detection results for the target

Findings

Version banners are revealed for each service (for example vsftpd 2.3.4, OpenSSH, Apache and Samba). Many of these versions are outdated and have publicly documented vulnerabilities.

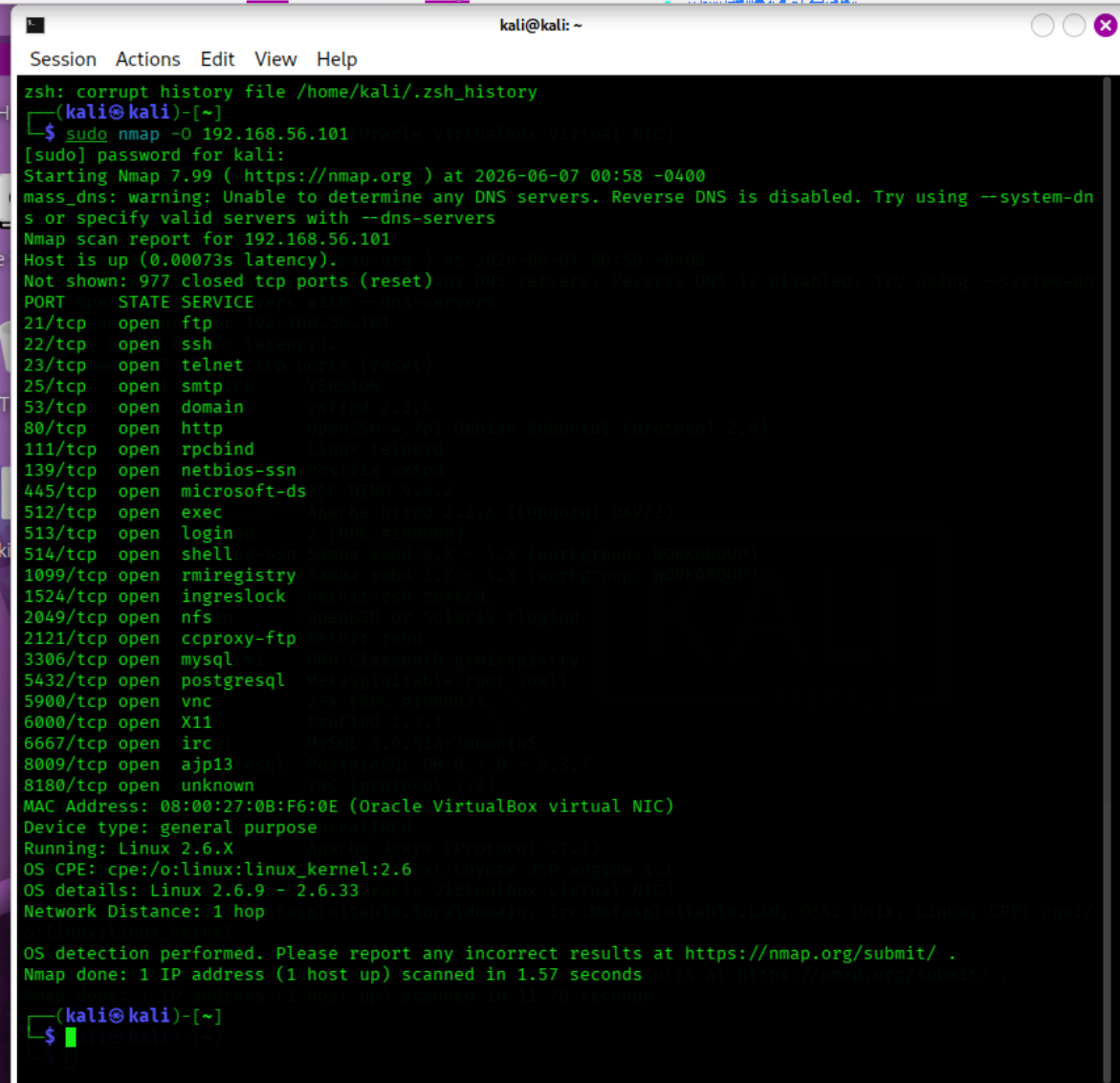
Challenge 17 — OS Detection

Command

```
sudo nmap -O 192.168.56.101
```

Explanation

The -O option uses TCP/IP stack fingerprinting to estimate the target operating system. The OS guess guides exploit selection, since most exploits are operating-system and kernel specific.



```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ sudo nmap -O 192.168.56.101  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-07 00:58 -0400  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.101  
Host is up (0.00073s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
MAC Address: 08:00:27:0B:F6:0E (Oracle VirtualBox virtual NIC)  
Device type: general purpose  
Running: Linux 2.6.X  
OS CPE: cpe:/o:linux:linux_kernel:2.6  
OS details: Linux 2.6.9 - 2.6.33  
Network Distance: 1 hop  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 1.57 seconds  
(kali@kali)-[~]  
$
```

Figure 5: Operating system fingerprinting result

Findings

Nmap identifies the host as Linux running a 2.6.x kernel, which is consistent with the Metasploitable 2 platform.

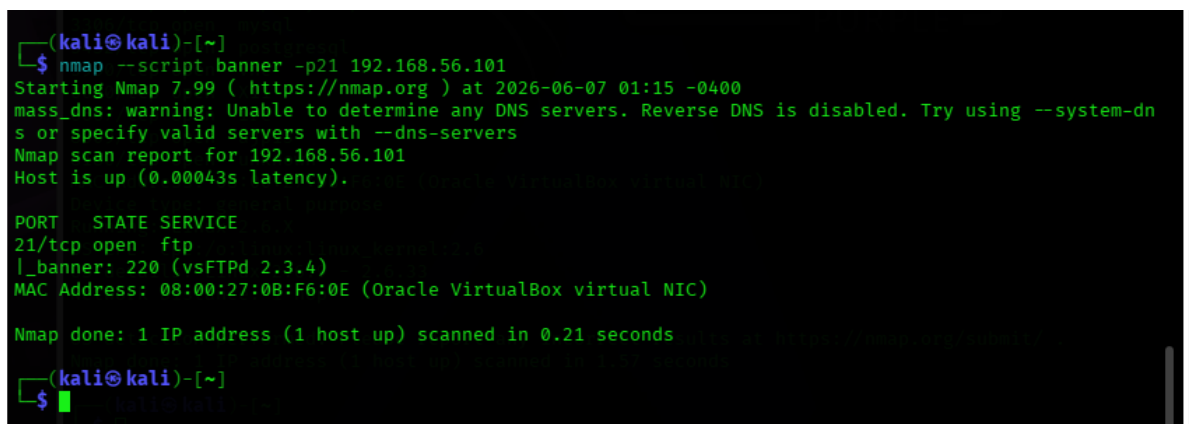
Challenge 9 — FTP Banner Grabbing

Command

```
nmap --script banner -p21 192.168.56.101
```

Explanation

Banner grabbing connects to a service and reads the welcome message it returns, which often discloses the exact software and version. For FTP this reveals the FTP daemon and its version number.



```
(kali㉿kali)-[~]
$ nmap --script banner -p21 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-07 01:15 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns
or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.101
Host is up (0.00043s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
_banner:  220 (vsFTPD 2.3.4)
MAC Address: 08:00:27:0B:F6:0E (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.21 seconds
```

Figure 6: FTP service banner showing the daemon version

Findings

The banner returns 220 (vsFTPD 2.3.4). This particular version is well known for shipping with a backdoor, making the FTP service a high-value target for exploitation.

Challenge 10 — Anonymous FTP Login

Command

```
ftp 192.168.56.101 (login as: anonymous)
```

Explanation

Misconfigured FTP servers often allow anonymous login, letting anyone connect without valid credentials and browse or download files. Testing for this checks whether unauthenticated file access is possible.



```
(kali㉿kali)-[~]
$ ftp 192.168.56.101
Connected to 192.168.56.101.
220 (vsFTPD 2.3.4)
Name (192.168.56.101:kali): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (||65027|).
150 Here comes the directory listing.
226 Directory send OK.
ftp>
```

Figure 7: Anonymous login attempt against the FTP service

Findings

The FTP service accepts the anonymous account, confirming weak access control that allows unauthenticated users to interact with the server.

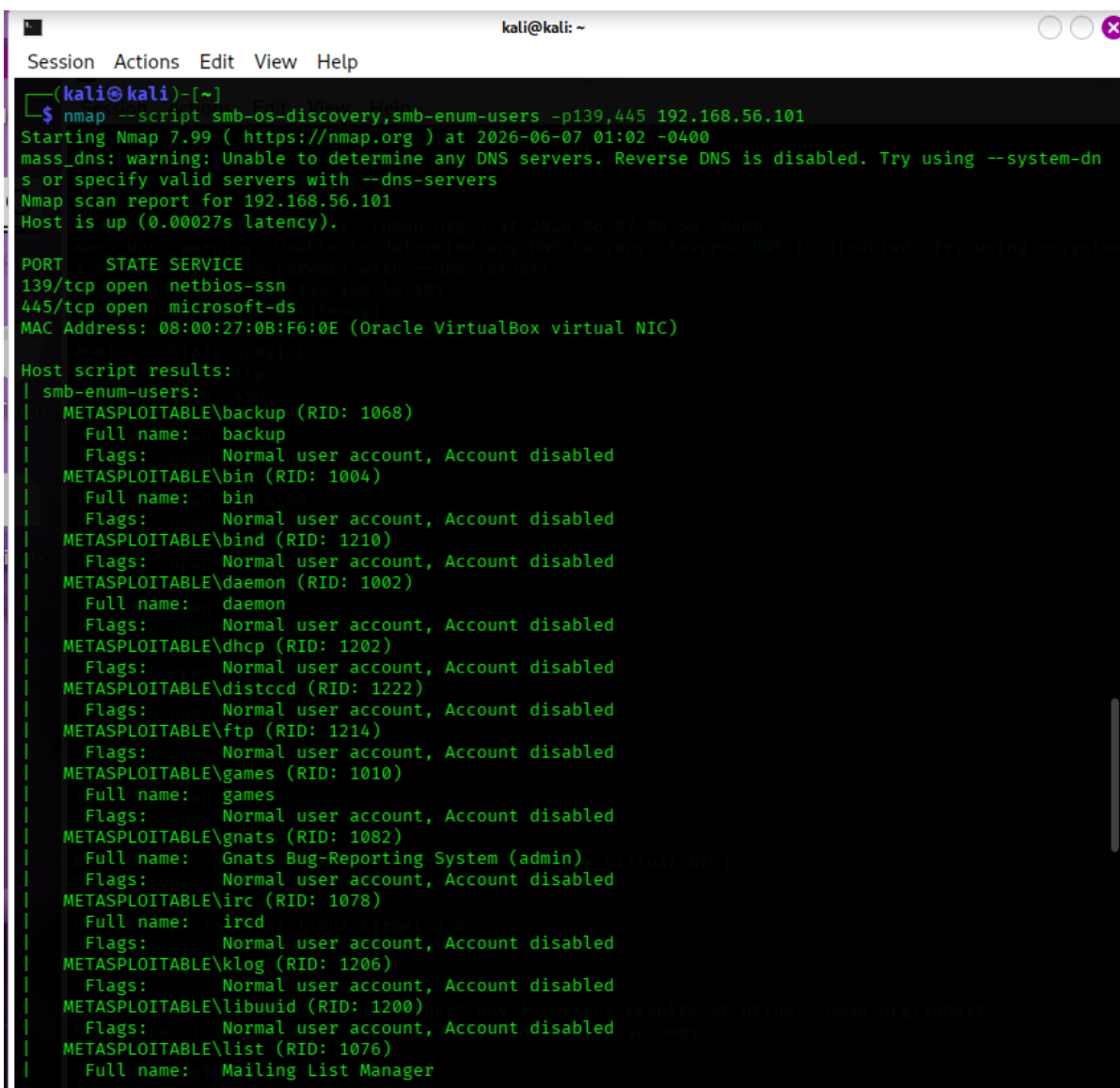
Challenge 11 — SMB NSE Enumeration

Command

```
nmap --script smb-os-discovery,smb-enum-users -p139,445 192.168.56.101
```

Explanation

SMB (ports 139 and 445) provides Windows-style file and printer sharing, implemented here through Samba. Nmap's SMB NSE scripts query the service through a null session to disclose the OS/Samba version and enumerate local user accounts.

A screenshot of a Kali Linux terminal window. The window title is 'kali@kali: ~'. The terminal shows the execution of the command 'nmap --script smb-os-discovery,smb-enum-users -p139,445 192.168.56.101'. The output includes the Nmap version (7.99), a warning about DNS, the scan report for 192.168.56.101, and a list of open ports (139/tcp for netbios-ssn and 445/tcp for microsoft-ds). The 'Host script results' section shows the output of the 'smb-enum-users' script, listing various user accounts on the remote system, including backup, bin, bind, daemon, dhcp, distccd, ftp, games, gnats, ircd, klog, libuuid, and list, along with their full names and flags.

```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali)~$ nmap --script smb-os-discovery,smb-enum-users -p139,445 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-07 01:02 -0400  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns  
or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.101  
Host is up (0.00027s latency).  
  
PORT      STATE SERVICE  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: 08:00:27:0B:F6:0E (Oracle VirtualBox virtual NIC)  
  
Host script results:  
| smb-enum-users:  
| METASPLOITABLE\backup (RID: 1068)  
|   Full name:    backup  
|   Flags:       Normal user account, Account disabled  
| METASPLOITABLE\bin (RID: 1004)  
|   Full name:    bin  
|   Flags:       Normal user account, Account disabled  
| METASPLOITABLE\bind (RID: 1210)  
|   Full name:    bind  
|   Flags:       Normal user account, Account disabled  
| METASPLOITABLE\daemon (RID: 1002)  
|   Full name:    daemon  
|   Flags:       Normal user account, Account disabled  
| METASPLOITABLE\dhcp (RID: 1202)  
|   Full name:    dhcp  
|   Flags:       Normal user account, Account disabled  
| METASPLOITABLE\distccd (RID: 1222)  
|   Full name:    distccd  
|   Flags:       Normal user account, Account disabled  
| METASPLOITABLE\ftp (RID: 1214)  
|   Full name:    ftp  
|   Flags:       Normal user account, Account disabled  
| METASPLOITABLE\games (RID: 1010)  
|   Full name:    games  
|   Flags:       Normal user account, Account disabled  
| METASPLOITABLE\gnats (RID: 1082)  
|   Full name:    Gnats Bug-Reporting System (admin)  
|   Flags:       Normal user account, Account disabled  
| METASPLOITABLE\irc (RID: 1078)  
|   Full name:    ircd  
|   Flags:       Normal user account, Account disabled  
| METASPLOITABLE\klog (RID: 1206)  
|   Full name:    klog  
|   Flags:       Normal user account, Account disabled  
| METASPLOITABLE\libuuid (RID: 1200)  
|   Full name:    libuuid  
|   Flags:       Normal user account, Account disabled  
| METASPLOITABLE\list (RID: 1076)  
|   Full name:    Mailing List Manager
```

Figure 8: SMB enumeration output (part 1)

```
kali@kali: ~
Session Actions Edit View Help
|
|  Flags:      Normal user account, Account disabled
| METASPLOITABLE\proftpd (RID: 1226)
|  Flags:      Normal user account, Account disabled
| METASPLOITABLE\proxy (RID: 1026)
|  Full name:  proxy
|  Flags:      Normal user account, Account disabled
| METASPLOITABLE\root (RID: 1000)
|  Full name:  root
|  Flags:      Normal user account, Account disabled
| METASPLOITABLE\service (RID: 3004)
|  Full name:  ,,,
|  Flags:      Normal user account, Account disabled
| METASPLOITABLE\sshd (RID: 1208)
|  Flags:      Normal user account, Account disabled
| METASPLOITABLE\sync (RID: 1008)
|  Full name:  sync
|  Flags:      Normal user account, Account disabled
| METASPLOITABLE\sys (RID: 1006)
|  Full name:  sys
|  Flags:      Normal user account, Account disabled
| METASPLOITABLE\syslog (RID: 1204)
|  Flags:      Normal user account, Account disabled
| METASPLOITABLE\telnetd (RID: 1224)
|  Flags:      Normal user account, Account disabled
| METASPLOITABLE\tomcat55 (RID: 1220)
|  Flags:      Normal user account, Account disabled
| METASPLOITABLE\user (RID: 3002)
|  Full name:  just a user,111,,
|  Flags:      Normal user account
| METASPLOITABLE\uucp (RID: 1020)
|  Full name:  uucp
|  Flags:      Normal user account, Account disabled
| METASPLOITABLE\www-data (RID: 1066)
|  Full name:  www-data
|  Flags:      Normal user account, Account disabled
|_ smb-os-discovery:
| OS: Unix (Samba 3.0.20-Debian)
| Computer name: metasploitable
| NetBIOS computer name:
| Domain name: localdomain
| FQDN: metasploitable.localdomain
|_ System time: 2026-06-07T01:02:22-04:00
Nmap done: 1 IP address (1 host up) scanned in 0.27 seconds
(kali@kali)-[~]
$
```

Figure 9: SMB user list and OS discovery (part 2)

Findings

smb-os-discovery reports the OS as Unix (Samba 3.0.20-Debian), computer name “metasploitable” and domain “localdomain”. smb-enum-users lists many valid accounts (root, user, msfadmin, sys, sshd and others), exposing usernames that can be reused in later password-guessing attacks.

Challenge 12 — SMB User & Share Enumeration (Enum4linux)

Command

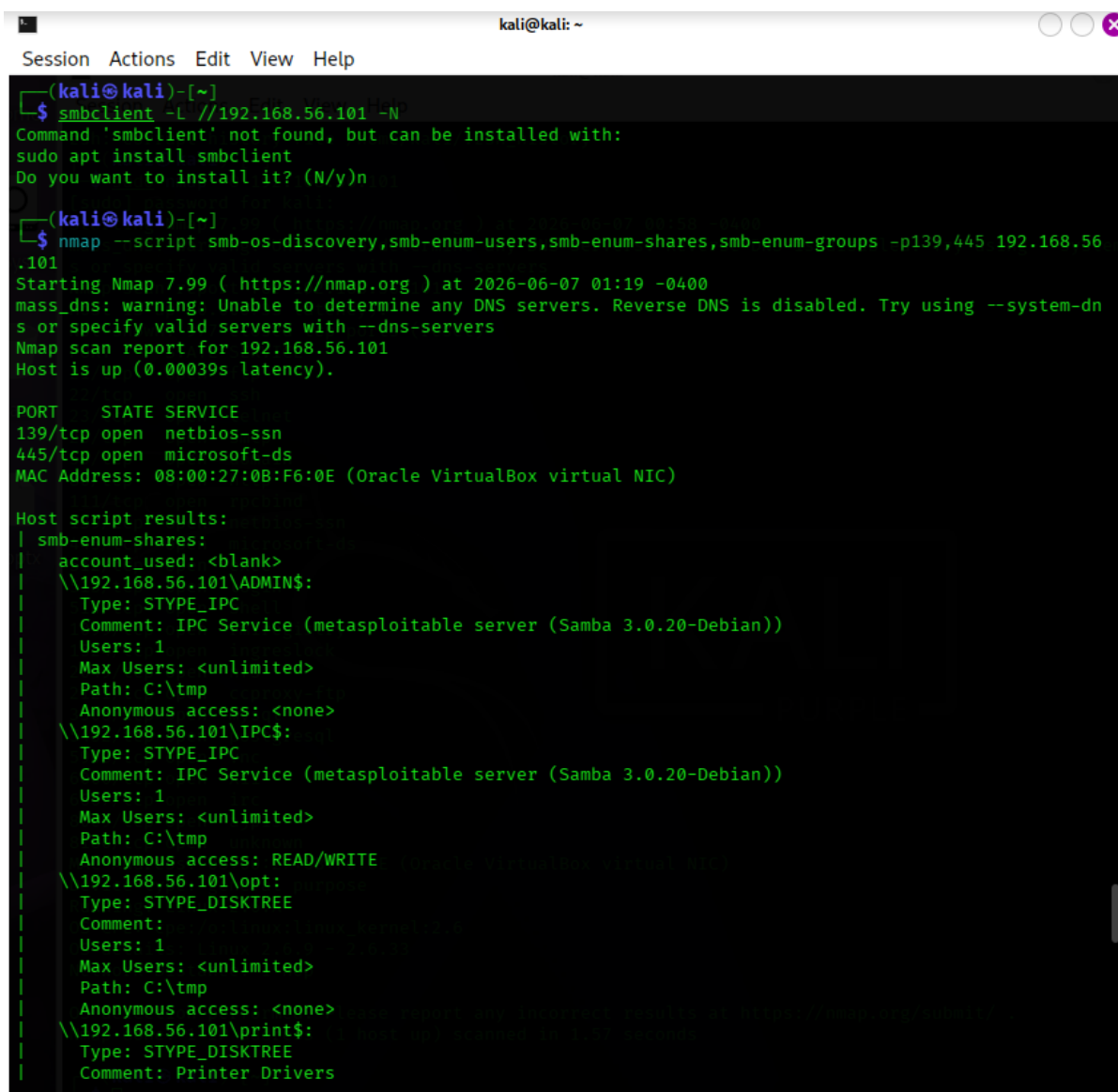
```
enum4linux -a 192.168.56.101
```

alternative used (tool not installed):

```
nmap --script smb-os-discovery,smb-enum-users,smb-enum-shares,smb-enum-groups -p139,445 192.168.56.101
```

Explanation

enum4linux automates SMB/NetBIOS enumeration to harvest users, groups, shares and the password policy in a single command. Where enum4linux is not installed, the equivalent Nmap SMB scripts collect the same information. This extends Challenge 11 by also listing shares and groups.



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ smbclient -L //192.168.56.101 -N  
Command 'smbclient' not found, but can be installed with:  
sudo apt install smbclient  
Do you want to install it? (N/y)n  
(kali@kali)-[~]  
$ nmap --script smb-os-discovery,smb-enum-users,smb-enum-shares,smb-enum-groups -p139,445 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-07 01:19 -0400  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.101  
Host is up (0.00039s latency).  
  
PORT      STATE SERVICE  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: 08:00:27:0B:F6:0E (Oracle VirtualBox virtual NIC)  
  
Host script results:  
| smb-enum-shares: |  
|   account_used: <blank> |  
|   \\192.168.56.101\ADMIN$: |  
|     Type: STYPE_IPC |  
|     Comment: IPC Service (metasploitable server (Samba 3.0.20-Debian)) |  
|     Users: 1 |  
|     Max Users: <unlimited> |  
|     Path: C:\tmp |  
|     Anonymous access: <none> |  
|   \\192.168.56.101\IPC$: |  
|     Type: STYPE_IPC |  
|     Comment: IPC Service (metasploitable server (Samba 3.0.20-Debian)) |  
|     Users: 1 |  
|     Max Users: <unlimited> |  
|     Path: C:\tmp |  
|     Anonymous access: READ/WRITE |  
|   \\192.168.56.101\opt: |  
|     Type: STYPE_DISKTREE |  
|     Comment: |  
|     Users: 1 |  
|     Max Users: <unlimited> |  
|     Path: C:\tmp |  
|     Anonymous access: <none> |  
|   \\192.168.56.101\print$: |  
|     Type: STYPE_DISKTREE |  
|     Comment: Printer Drivers |
```

Figure 10: SMB shares and groups enumerated from the target

Findings

The available shares (such as IPC\$, print\$, tmp and opt) and group memberships are enumerated, mapping which resources can be reached over SMB without authentication.

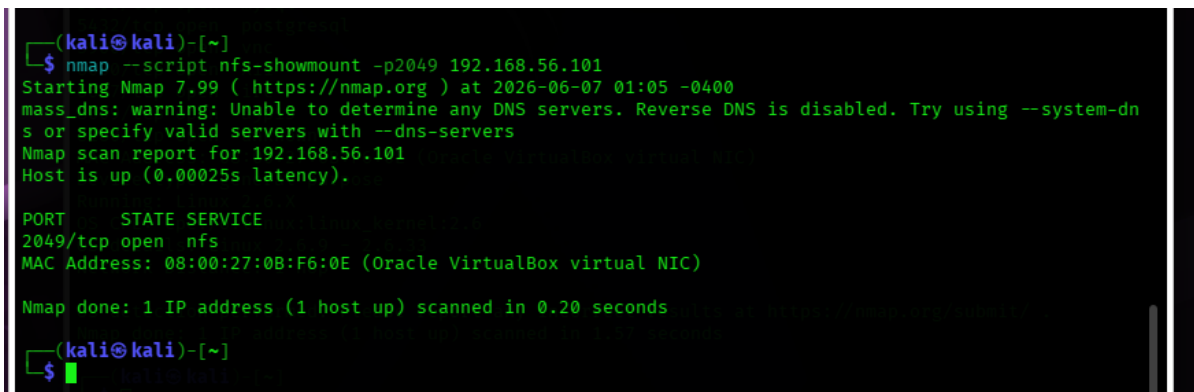
Challenge 13 — NFS Exports

Command

```
nmap --script nfs-showmount -p2049 192.168.56.101
```

Explanation

NFS (port 2049) lets clients mount remote directories over the network as if they were local. Enumerating the service reveals whether NFS is available and which directories are exported — a misconfigured export can give direct, unauthenticated file access.



```
(kali㉿kali)-[~]
$ nmap --script nfs-showmount -p2049 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-07 01:05 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns
or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.101
Host is up (0.00025s latency).

PORT      STATE SERVICE
2049/tcp  open  nfs
MAC Address: 08:00:27:0B:F6:0E (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds
```

Figure 11: NFS service detected on port 2049

Findings

NFS is confirmed open on TCP port 2049. The export list is served through the mountd service (see Challenge 19); together these confirm that remote file sharing is exposed on the host.

Challenge 19 — RPC Info

Command

```
rpcinfo -p 192.168.56.101
```

Explanation

rpcinfo queries the RPC portmapper on port 111 to list every RPC-based service and the port it listens on. This exposes services such as mountd and NFS that may not be obvious from a normal port scan.

```
(kali@kali)-[~]
$ rpcinfo -p 192.168.56.101
program vers proto  port  service
100000    2    tcp    111   portmapper
100000    2    udp    111   portmapper
100024    1    udp    46147 status
100024    1    tcp    34488 status
100003    2    udp    2049  nfs
100003    3    udp    2049  nfs
100003    4    udp    2049  nfs
100021    1    udp    37722 nlockmgr
100021    3    udp    37722 nlockmgr
100021    4    udp    37722 nlockmgr
100003    2    tcp    2049  nfs
100003    3    tcp    2049  nfs
100003    4    tcp    2049  nfs
100021    1    tcp    58568 nlockmgr
100021    3    tcp    58568 nlockmgr
100021    4    tcp    58568 nlockmgr
100005    1    udp    60298 mountd
100005    1    tcp    52793 mountd
100005    2    udp    60298 mountd
100005    2    tcp    52793 mountd
100005    3    udp    60298 mountd
100005    3    tcp    52793 mountd
```

Figure 12: RPC services registered with the portmapper

Findings

The portmapper exposes portmapper (111), status, nfs (2049), nlockmgr and mountd. This confirms an active NFS/RPC stack and directly supports the NFS findings above.

Challenge 7 — SMTP VRFY / EXPN

Command

```
nmap -p25 --script smtp-enum-users,smtp-commands 192.168.56.101
```

Explanation

SMTP servers may support the VRFY and EXPN commands, which confirm whether a username or mailbox exists. If enabled, these commands can be abused to enumerate valid users. Listing the supported commands shows whether this weakness is present.

```

(kali@kali)-[~]
$ nmap -p25 --script smtp-enum-users,smtp-commands 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-07 01:16 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns
or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.101
Host is up (0.00028s latency).

PORT      STATE SERVICE
25/tcp    open  smtp
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCED
STATUSCODES, 8BITMIME, DSN
|_smtp-enum-users:
|_ Method RCPT returned a unhandled status code.
MAC Address: 08:00:27:0B:F6:0E (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.22 seconds

```

Figure 13: SMTP supported commands, showing VRFY is enabled

Findings

The server (metasploitable.localdomain) advertises VRFY among its supported SMTP commands, meaning it can be used to validate usernames. The internal hostname is also leaked in the response.

4.0 Summary of Attack Surface

The table below summarises the key findings from the enumeration phase and the technique used to obtain each one.

Service / Port	Technique	Key Finding
FTP (21)	Banner grab / anon login	vsftpd 2.3.4 (backdoored); anonymous login allowed
SMB (139/445)	NSE / enum4linux	Samba 3.0.20; users, shares & groups disclosed via null session
NFS (2049) + RPC (111)	nfs-showmount / rpcinfo	NFS, mountd and nlockmgr exposed; remote file sharing available
SMTP (25)	VRFY / EXPN	VRFY enabled — allows username enumeration
Host (general)	nmap -sV / -O	Linux 2.6.x kernel; many outdated, vulnerable services

5.0 Conclusion

Enumeration of the target revealed a very large attack surface. The host runs numerous outdated services, several of which (most notably vsftpd 2.3.4) carry publicly known vulnerabilities. Access control is weak: the FTP service permits anonymous login, the SMB service discloses users, groups and shares through a null session, and NFS/RPC services

expose remote file sharing. The SMTP service additionally allows username enumeration through the VRFY command.

Collectively these findings provide multiple concrete footholds for the exploitation phase. To reduce the attack surface, the operator should update all services to supported versions, disable anonymous FTP, restrict SMB null sessions and unnecessary shares, lock down NFS exports to trusted hosts only, and disable the SMTP VRFY/EXPN commands.